



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	STA4001
Course title (English)	Stochastic Processes
Course title (Chinese)	随机过程
Units	3
Language of Instruction	English
Description (English)	This course covers basic stochastic models as well as their various applications including service systems, financial engineering, etc. Through these stochastic models, we will learn quantitative methods which are useful in analyzing, designing, and operating one of these systems. Topics to be covered include 1. Review of probability theory 2. Discrete time Markov chains 3. Poisson processes 4. Continuous time Markov chains and queuing model 5. Brownian motion.
Description (Chinese)	本课程涵盖基本的随机模型及其各种应用，包括服务系统、金融工程等。通过这些随机模型，我们将学习可用于分析、设计和操作其中一个系统的定量方法。 将涵盖的主题包括且不限于： 1. 概率论复习 2. 离散时间马尔可夫链 3. 泊松过程 4. 连续时间马尔可夫链和排队论 5. 布朗运动。

2. Prerequisites / Co-requisites

A. Prerequisites

(MAT1002 or MAT1002B or MAT1004 or MAT1012) and (MAT2040 or MAT2041 or MAT2041A or MAT2042) and (ECO2121 or STA2001 or STA2001H or STA2003)

B. Co-requisites



None

3. Learning Outcomes

Upon completing this course, students are expected to

1. Understand the concepts of discrete time and continuous time Markov chains
2. Master the technical skills for transient and steady-state analysis of Markov chains through a combination of analytical, numerical, and Monte Carlo methods.
3. Be able to analyze and simulate (time-homogeneous and non-homogeneous) Poisson processes.
4. Learn how to use Markov chains to model inventory systems, communications networks, service systems, and financial systems.
5. Be able to use Markov chains to conduct performance analysis of various service and engineering systems.
6. Understand the concept of Brownian motions and basic stochastic calculus.

4. Course syllabus

This course covers basic stochastic models and their various applications including inventory management, information systems, manufacturing and service systems. Through these stochastic models, we will learn quantitative methods which are useful in analyzing, designing, and operating one of these systems.

Topics to be covered include

1. Review of probability theory
2. Discrete time Markov chains
3. Poisson processes
4. Continuous time Markov chains and queuing model
5. Brownian motion.

5. Assessment Scheme

Component/ method	% weight
Coursework (including homework, quizzes, etc)	30
Mid-Term Exam	30
Final Exam	40



6. Grade descriptor

General

Grade	Description
A	Outstanding performance on learning outcomes 1-6.
A-	Generally outstanding performance on learning outcomes 1-5, substantial performance on learning outcomes 6.
B	Substantial performance on learning outcomes 1-4, acceptable performance on learning outcomes 5-6.
C	Satisfactory performance on learning outcomes 1-3, acceptable performance on learning outcomes 4-6.
D	Barely satisfactory performance on learning outcomes 1-3
F	Failure to achieve barely satisfactory performance on learning outcomes 1-3

7. Feedback for evaluation

Formal Course and Teaching Evaluation
Informal feedback and/or teaching assistant
School retreat and program review

8. Reading

Required Reading

1.Introduction to Probability Models, 9780123756862, Sheldon M. Ross, Academic Press., 11TH EDITION, 2014, United States

9. Course components

Activity	Hours/week
Lecture	3
Tutorial	1

10. Indicative teaching plan



Week	Content/ topic/ activity
1	Conditional probability and conditional expectation
2	Conditional probability and conditional expectation
3	Discrete time Markov chains
4	Discrete time Markov chains
5	Discrete time Markov chains
6	Poisson processes
7	Poisson processes
8	Poisson processes
9	Continuous time Markov chains and queuing theory
10	Continuous time Markov chains and queuing theory
11	Continuous time Markov chains and queuing theory
12	Brownian motion
13	Brownian motion
14	Brownian motion

11. Any other information

12. Version date

Version number	7
As of (date)	2025/07/21